Kavimalar D 241801118 exp 6a

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the Netflix dataset
df = pd.read_csv('/content/netflix_titles.csv')

# Explore the dataset
print("First 5 rows:")
display(df.head())

print("\nLast 5 rows:")
display(df.tail())

print("\nDataset Shape:", df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

print("\nData Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Rows:", df.duplicated().sum())

print("\nStatistical Summary:")
display(df.describe(include='all'))
```

First 5 rows:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	descri
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her nea enc life, f
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	cr patl party, a To
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To prot family pc dru
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	I flirtatio toilet down
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a co c kn tr

Last 5 rows:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	desc
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007	R	158 min	Cult Movies, Dramas, Thrillers	A car rep
											Kids' TV	W

```
# Movies vs TV Shows
plt.figure(figsize=(7, 5))
sns.countplot(data=df, x='type')
plt.title('Movies vs TV Shows on Netflix')
plt.xlabel('Content Type')
plt.ylabel('Number of Titles')
plt.show()

print("Content Type Counts:")
display(df['type'].value_counts())

# Rating distribution
plt.figure(figsize=(12, 6))
sns.countplot(
    data=df,
    y='rating',
    order=df['rating'].value_counts().index
)
plt.title('Distribution of Netflix Ratings')
plt.xlabel('Number of Titles')
plt.ylabel('Rating')
plt.show()

# Top 10 countries
countries = df['country'].dropna().str.split(', ').explode()
top_countries = countries.value_counts().head(10)

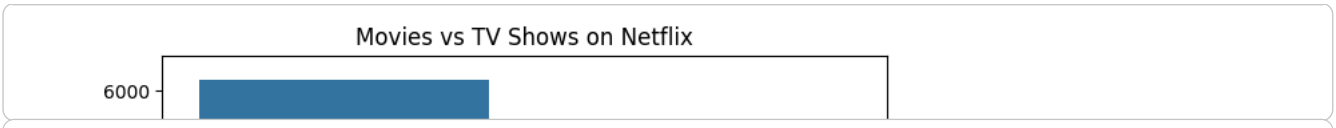
plt.figure(figsize=(12, 6))
top_countries.sort_values().plot(kind='barh')
plt.title('Top 10 Countries Producing Netflix Content')
plt.xlabel('Number of Titles')
plt.ylabel('Country')
plt.show()

print("Top 10 Countries:")
display(top_countries)
```

duration object

```
listed_in       object
description     object
dtype: object

Missing Values:
show_id            0
type               0
title              0
director        2634
cast             825
country          831
date_added        10
release_year       0
```




```
# Convert date columns to datetime
df['date_added'] = pd.to_datetime(df['date_added'].str.strip(), errors='coerce')

# Release year trend
release_counts = df['release_year'].value_counts().sort_index()

plt.figure(figsize=(12, 5))
release_counts.plot()
plt.title('Netflix Titles by Release Year')
plt.xlabel('Release Year')
plt.ylabel('Number of Titles')
plt.show()

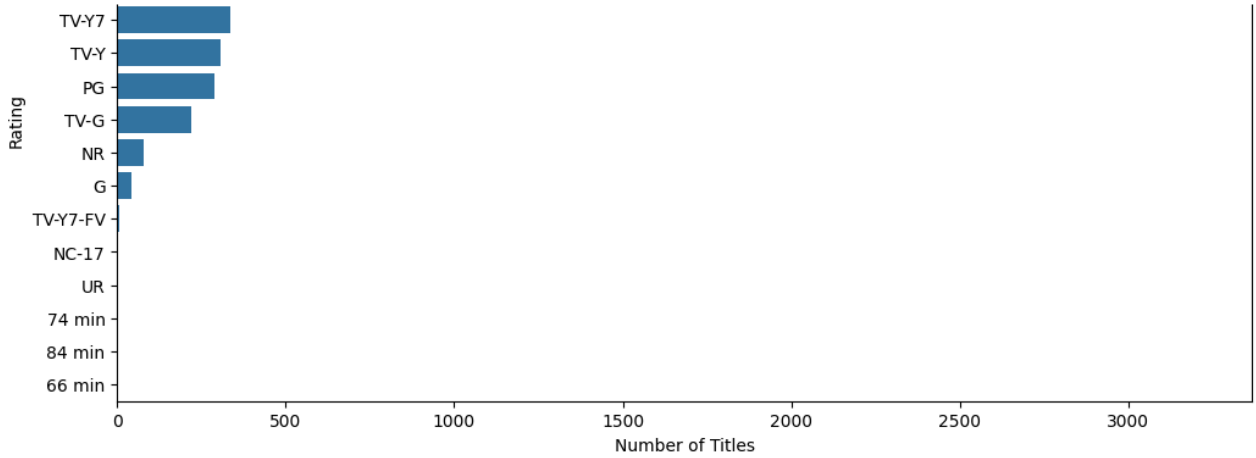
print("Top 10 Release Years:")
display(df['release_year'].value_counts().head(10))

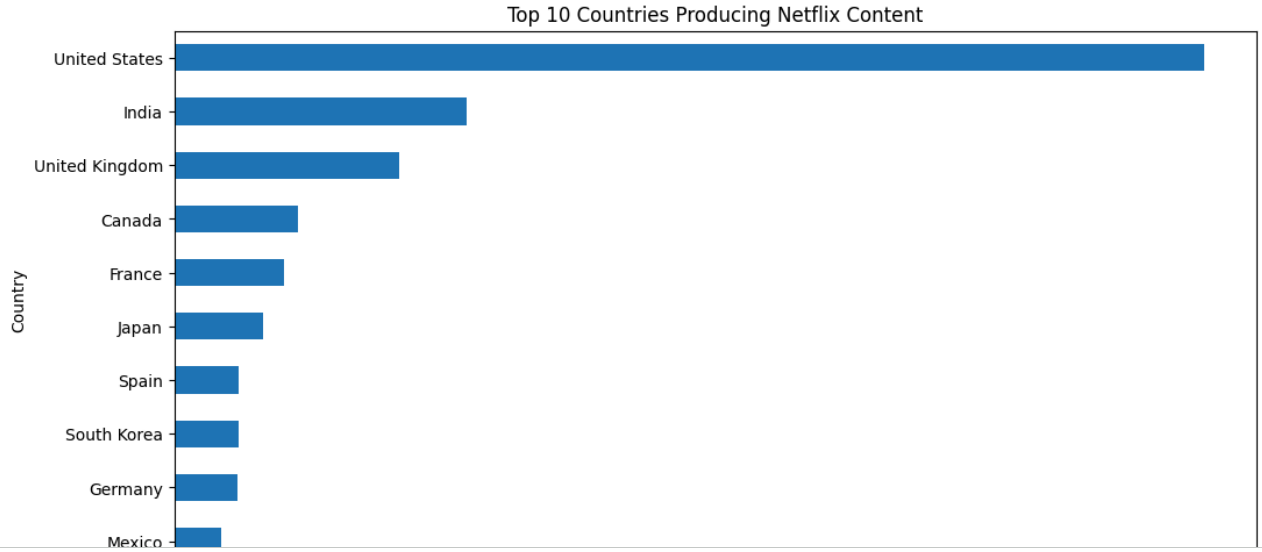
# Titles added to Netflix by year
df['year_added'] = df['date_added'].dt.year
added_counts = df['year_added'].value_counts().sort_index()

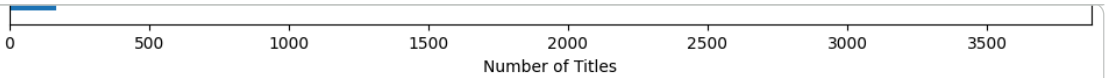
plt.figure(figsize=(12, 5))
added_counts.plot(kind='bar')
plt.title('Netflix Titles Added by Year')
plt.xlabel('Year Added')
plt.ylabel('Number of Titles')
plt.show()

print("Top 10 Years by Titles Added:")
display(df['year_added'].value_counts().head(10))

# Content growth trend over time
plt.figure(figsize=(12, 5))
added_counts.sort_index().cumsum().plot()
plt.title('Cumulative Netflix Content Addition Trend')
plt.xlabel('Year')
plt.ylabel('Cumulative Titles Added')
plt.show()
```







Top 10 Countries:

country	count
United States	3689
India	1046
United Kingdom	804
Canada	445
France	393
Japan	318
Spain	232
South Korea	231
Germany	226
Mexico	169

dtype: int64




```
# Extract movie durations in minutes
movies = df[df['type'] == 'Movie'].copy()
movies['duration_minutes'] = pd.to_numeric(
    movies['duration'].str.extract(r'(\d+)')[0],
    errors='coerce'
)

plt.figure(figsize=(10, 5))
sns.histplot(movies['duration_minutes'].dropna(), bins=30, kde=True)
plt.title('Distribution of Movie Durations')
plt.xlabel('Duration (Minutes)')
plt.ylabel('Number of Movies')
plt.show()

print("Movie Duration Statistics:")
display(movies['duration_minutes'].describe())

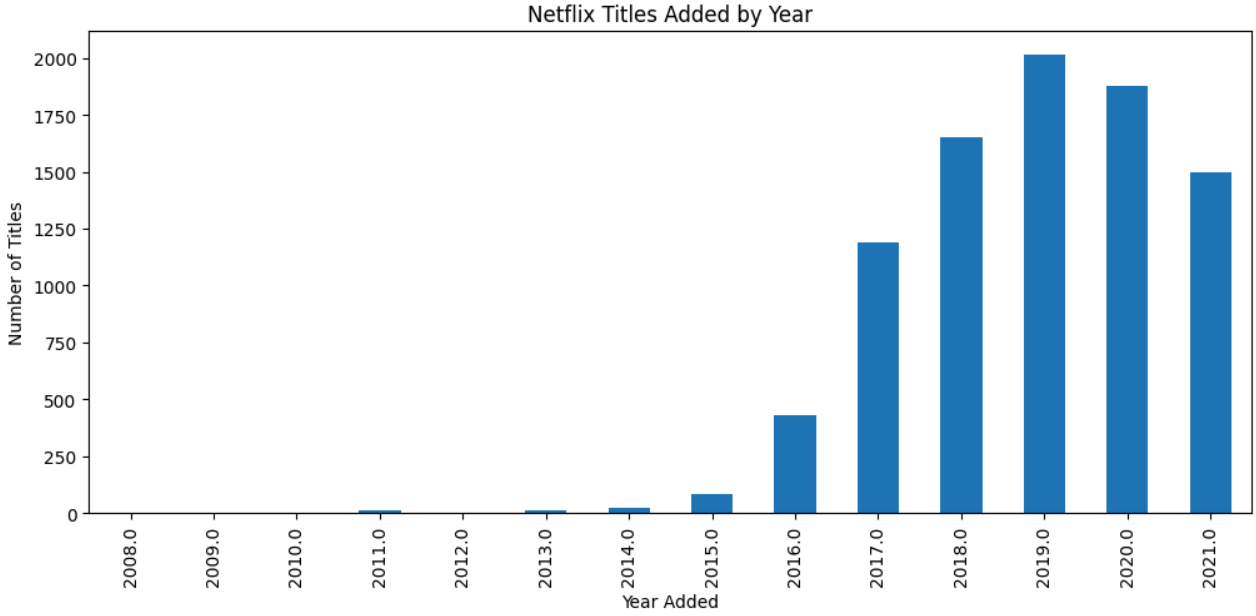
# Extract TV show seasons
tv_shows = df[df['type'] == 'TV Show'].copy()
tv_shows['seasons'] = pd.to_numeric(
    tv_shows['duration'].str.extract(r'(\d+)')[0],
    errors='coerce'
)

plt.figure(figsize=(10, 5))
season_counts = tv_shows['seasons'].value_counts().sort_index()
season_counts.plot(kind='bar')
plt.title('Number of TV Shows by Seasons')
plt.xlabel('Number of Seasons')
plt.ylabel('Number of TV Shows')
plt.show()

print("TV Shows by Number of Seasons:")
display(season_counts.sort_values(ascending=False))
```

2012	237

dtype: int64

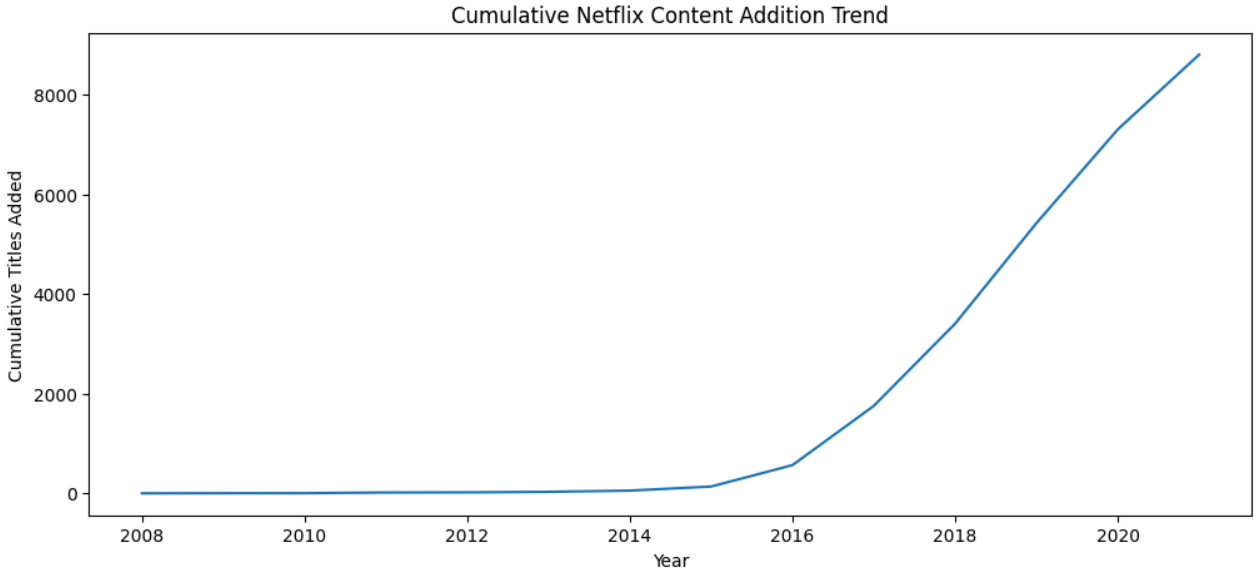



```
Top 10 Years by Titles Added:
```

year_added	count
2019.0	2016
2020.0	1879
2018.0	1649
2021.0	1498
2017.0	1188

2016.0	429
2015.0	82
2014.0	24
2011.0	13
2013.0	11

dtype: int64



```
# Top 10 most common genres
genres = df['listed_in'].dropna().str.split(', ').explode()
top_genres = genres.value_counts().head(10)

plt.figure(figsize=(12, 6))
top_genres.sort_values().plot(kind='barh')
plt.title('Top 10 Most Common Netflix Genres')
plt.xlabel('Number of Titles')
plt.ylabel('Genre')
plt.show()

print("Top 10 Genres:")
display(top_genres)

# Top 10 directors
directors = df['director'].dropna().str.split(', ').explode()
top_directors = directors.value_counts().head(10)

plt.figure(figsize=(12, 6))
top_directors.sort_values().plot(kind='barh')
plt.title('Top 10 Directors on Netflix')
plt.xlabel('Number of Titles')
plt.ylabel('Director')
plt.show()

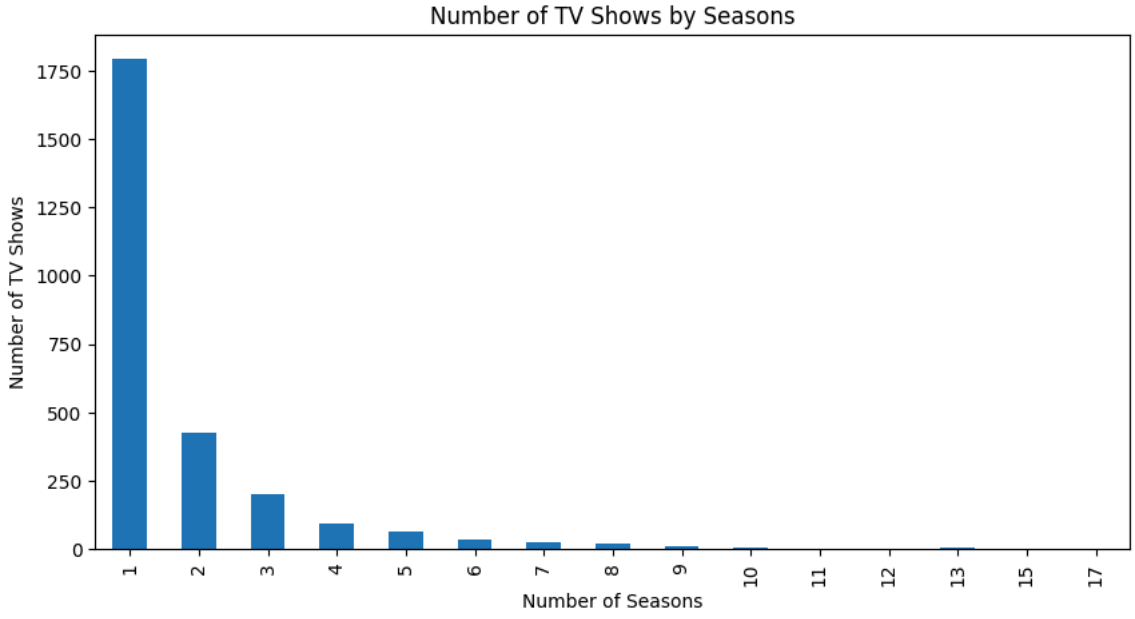
print("Top 10 Directors:")
display(top_directors)

# Top 10 actors
actors = df['cast'].dropna().str.split(', ').explode()
top_actors = actors.value_counts().head(10)

plt.figure(figsize=(12, 6))
top_actors.sort_values().plot(kind='barh')
plt.title('Top 10 Actors in Netflix Content')
plt.xlabel('Number of Titles')
plt.ylabel('Actor')
plt.show()

print("Top 10 Actors:")
display(top_actors)
```

dtype: float64

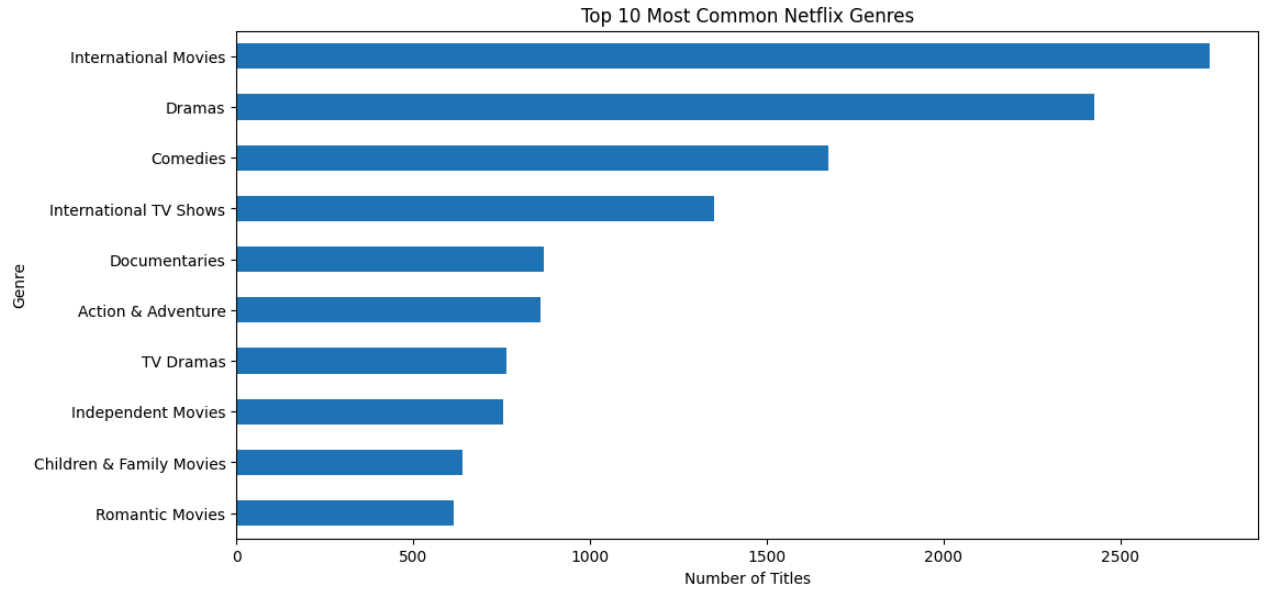


TV Shows by Number of Seasons:

seasons	count
1	1793
2	425
3	199
4	95
5	65
6	33
7	23

7	25
8	17
9	9
10	7
13	3
11	2
12	2
15	2
17	1

dtype: int64



Top 10 Genres:

listed_in	count
International Movies	2752

```
!pip install -q wordcloud

from wordcloud import WordCloud, STOPWORDS

# Combine all descriptions
text = ' '.join(df['description'].dropna().astype(str))

# Generate word cloud
wordcloud = WordCloud(
    width=1200,
    height=600,
    background_color='white',
    stopwords=STOPWORDS,
    max_words=100
).generate(text)

plt.figure(figsize=(14, 7))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Most Common Words in Netflix Descriptions')
plt.show()

# Netflix content additions by month
df['month_added'] = df['date_added'].dt.month

month_names = {
    1: 'January', 2: 'February', 3: 'March',
    4: 'April', 5: 'May', 6: 'June',
    7: 'July', 8: 'August', 9: 'September',
    10: 'October', 11: 'November', 12: 'December'
}

monthly_counts = df['month_added'].value_counts().sort_index()
monthly_counts.index = monthly_counts.index.map(month_names)

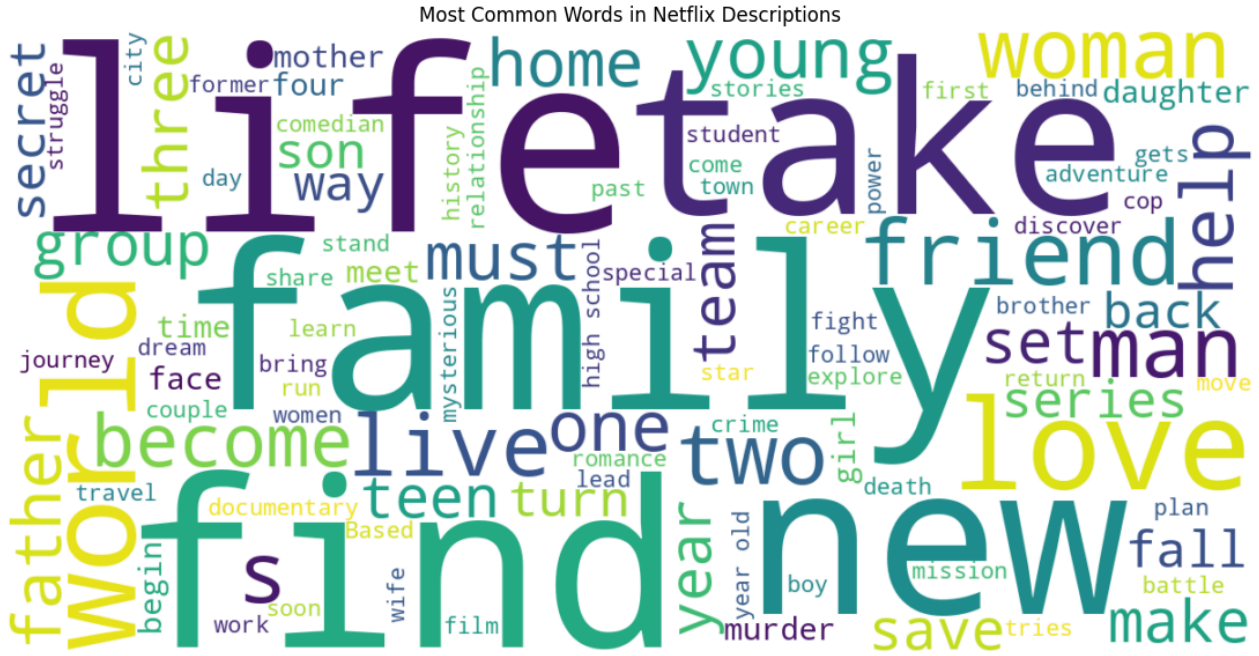
plt.figure(figsize=(12, 5))
monthly_counts.plot(kind='bar')
plt.title('Netflix Content Additions by Month')
plt.xlabel('Month')
plt.ylabel('Number of Titles Added')
plt.xticks(rotation=45)
plt.show()

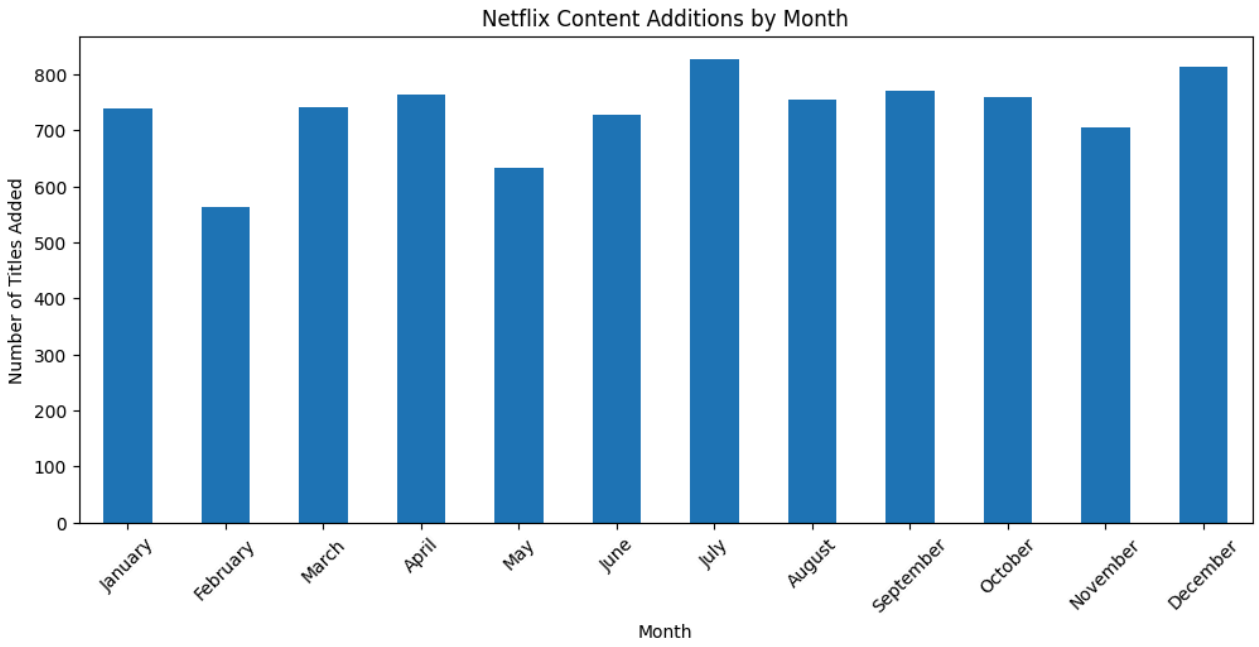
print("Content Additions by Month:")
display(monthly_counts.sort_values(ascending=False))
```

-

Jan Suter	21
Raúl Campos	19
Suhas Kadav	16

Marcus Raboy 16





Content Additions by Month:

Yuki Kaji	29
count	
Amitabh Bachchan	28
month_added	
July	827

```
# 1. Countries producing more Movies vs TV Shows
country_type = df.dropna(subset=['country']).copy()
country_type['country'] = country_type['country'].str.split(', ')
country_type = country_type.explode('country')

top_countries = country_type['country'].value_counts().head(10).index
country_type = country_type[country_type['country'].isin(top_countries)]

country_pivot = pd.crosstab(
    country_type['country'], country_type['type']
)

country_pivot.plot(kind='bar', figsize=(12, 6))
plt.title('Movies vs TV Shows by Top Countries')
plt.xlabel('Country')
plt.ylabel('Number of Titles')
plt.xticks(rotation=45)
plt.show()

# 2. Most common ratings for Movies vs TV Shows
plt.figure(figsize=(12, 6))
sns.countplot(
    data=df.dropna(subset=['rating']),
```

```
        x='rating',
        hue='type',
        order=df['rating'].value_counts().index
    )
    plt.title('Ratings by Content Type')
    plt.xlabel('Rating')
    plt.ylabel('Number of Titles')
    plt.xticks(rotation=45)
    plt.show()

    print("Ratings by Content Type:")
    display(pd.crosstab(df['rating'], df['type']))

    # 3. Oldest and newest titles by release year
    valid_years = df.dropna(subset=['release_year'])

    print("Oldest title(s):")
    display(valid_years[valid_years['release_year'] ==
                        valid_years['release_year'].min()]
            [['title', 'type', 'release_year']])

    print("Newest title(s):")
    display(valid_years[valid_years['release_year'] ==
                        valid_years['release_year'].max()]
            [['title', 'type', 'release_year']])

    # 4. Genre co-occurrence heatmap
    from collections import Counter
    from itertools import combinations

    genre_pairs = Counter()

    for genres in df['listed_in'].dropna():
        genre_list = sorted(set(g.strip() for g in genres.split(',')))
        for g1, g2 in combinations(genre_list, 2):
            genre_pairs[(g1, g2)] += 1

    top_pairs = genre_pairs.most_common(15)

    if top_pairs:
        pair_df = pd.DataFrame(
            [(g1, g2, count) for (g1, g2), count in top_pairs],
            columns=['Genre 1', 'Genre 2', 'Count']
        )

        heatmap_data = pair_df.pivot(
            index='Genre 1', columns='Genre 2', values='Count'
        ).fillna(0)

        plt.figure(figsize=(12, 7))
        sns.heatmap(heatmap_data, annot=True, fmt='.0f', cmap='YlGnBu')
        plt.title('Top Genre Co-occurrence Heatmap')
        plt.show()

        print("Most Common Genre Pairs:")
        display(pair_df)
```